E350 Three Phase WC

E350 U3400

Fact Sheet

The second generation of intelligent endpoint, E350 is now available and ready to become a key component of your smart grid.

The E350 with more processor power and increased memory will help to future proof the meter as your AMI requirements evolve. The sheer power of the processor enables the meter to carry out the complex AMI metering functionality, decode ripple control signals and communicate locally and remotely to the Utilities AMI communication systems.

Date: 4/04/2016
Filename: E350 U3400 Fact Sheet V1.1

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E350 U3400 - Technical specifications

Functionality

PredicTemp ™

The meter supports a patented logic and algorithms to detect and alarm abnormal temperature rise with the meter installation.
This feature reduces the risk of potential fire hazards due to:

- Faulty installation (incorrect torque applied on terminal screws)
- Overload of the circuit

Demand Management

Load Control Relay State Machine

The meter features a comprehensive and unique state machine which handles different control sources simultaneously according a clearly defined hierarchy.
Currently the following control sources are supported (future extension planned):

- Loss of Supply or Disconnect
- Emergency Control (Under Frequency Load Control)
- Boost (primacy enabled/disabled)
- Priority override (AMI or local commands)
- Ripple control
- Time Table (internal load control command)

Measurement

Energy	
Energy Type	kWh, kVArh, kVAh
Registers	6
Demand	
Demand Type	kW, kVAr, kVA
Registers	2
TOU	
Rates	6 (A,B,C,D,E,F)
Switch time	15minute intervall
Rolling day snapshot	8
End of billing period snapshot	14

Profiles

Load Profile	
Number of channels	4
Number of sets	4
Interval (minutes)	10,15,30,60
Max days (30min, 2 channels)	1000

Event Log and QoS

Event Management	stores up to 500 events

Quality of Supply Monitoring

Voltage and current registers
Power fail counters
Under and over voltage events (Sag and Swell)
Power factor register
Load Profile of instantaneues quantities in the form of min, max, average, instantaneous

Fraud detection

Reverse energy
On/Off power tamper detection on terminal cover
Load side voltage detection

Ripple Control

Protocols

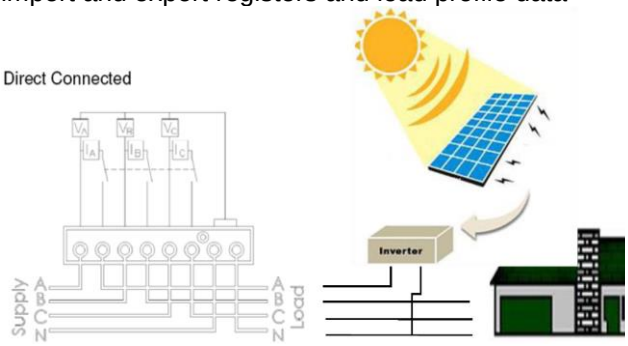
Pulsadis (Enertec)
Decabit (Zellweger)
Semagyr (Landis & Gyr)
Ricontic (Brown Boveri)

Standard compliance	AS 62054.11

Gross Metering

Overview

Gross metering for PV and cogeneration applications based on element 1 and element 2 import and export registers and load profile data



Multifunction Output Contacts

Pulsed outputs and Threshold Control

The meter supports up to 4 pairs of potential free output contacts which can are programmable to either pulsed outputs or threshold contacts.

Pulsed outputs

Energy, End of interval, TOU rate change

Threshold control

Under Frequency, Low Voltage, Over Demand

Four (4) sets of 2-wire potential free contacts
Voltage Rating: 40VAC max and 12V- 24VDC
Current Rating: 100 mA (DC or AC)
Relay type: Solid State Relay

Communcation Interface

Local Communication

Meter readable using Itron hand held units and MVRS software.

Baud rates	1200 to 38400

Level of passwords	three levels
Standard compliance	AS 1284.10.2

AMI Communication

Field upgradeable communications modules for various applications:
Landis+Gyr E35C-B Gridstream RF with ZigBee
Landis+Gyr E35C-3G Gridstream Module with Zigbee
Landis+Gyr E35C-WM Wimax Module with ZigBee
SSN RF Mesh with ZigBee
AMR Transparent modem CSD/PSD

Breaker

Disconnect Breaker

Built in 100A Disconnect Relay. It can be utilized for various functions such as individual premise disconnect or demand limiting applications.

Voltage	$240V_{AC}$
Max. current	100A

Control sources:

- AMI broadcast commands
- supply capacity control

Relay

Load Control Relay	
Optional a fully integrated 31.5A relay	
type	Bi-stable latching relay
voltage	$240V_{AC}$
max. current	31.5A

General

Measured Energy
kWh imported per phase and total
kWh exported per phase and total
kVARh imported per phase and total (Q1+Q2)
kVARh exported per phase and total (Q3+Q4)

Accuracy	AS62053.21
Wh	Class 1
VARh	Class 2

Clock	
Internal clock	accuracy < 2.5ppm
Mains based	AS 62054-21

Operating Range	
Voltage (phase to neutral)	$240V_{AC}$ ± 20%
Starting current	10(100)A, 40mA
Frequency	49-51Hz
Voltage burden at 240V	<10 VA, < 2W
Current burden	0.07VA

Absolute Maximum Ratings	
Current, continuous	100A
Current, short duration	3000A for 0.5cycle
Voltage, continuous	$288V_{AC}$
Insulation withstand	4kV for 1 minute
Impulse voltage withstand	12kV

Battery
Internal battery for time keeping and tamper support

Impermeability to AS60529	IP53

Temperature	
Specified operating range	-10 °C to +60°C
Limit range of operation	-20 °C to +70°C
Storage	-25 °C to +70°C

Humidity	Up to 95% non-condensing

Hardware	
Terminal cable size range	4 - 35mm^2
Weight	approx. 1.6 kg
Mounting holes	SDM
Terminal screws	slotted

Dimensions
Standard terminal cover – 229x175x109mm (h,w,d)
Extended terminal cover – 271x175x109mm (h,w,d)

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Standards

The E350 U3400 meter complies in full or in part with the following standards

AS 62052.11
Electricity metering equipment (AC) – General requirements, tests and test conditions - Part 11: Metering equipment

AS 62052.21
Electricity metering equipment (a.c.) - General requirements, tests and test conditions - Part 21: Tariff and load control equipment

AS 62053.21
Electricity metering equipment (a.c.) - Particular requirements - Part 21: Static meters for active energy (classes 1 and 2)

AS 62053.23
Electricity metering equipment (ac) - Particular requirements - Static meters for reactive energy (classes 2 and 3)

AS 62054.11
Electricity metering (ac) - Tariff and load control - Particular requirements for electronic ripple control receivers

AS 62054.21
Electricity metering (ac) - Tariff and load control - Particular requirements for time switches

IEC 62053-31
Electricity metering equipment (a.c.) - Particular requirements - Part 31: Pulse output devices for electromechanical and electronic meters (two wires only) and test certificates supplied

AS 1284.10.2
Data exchange for meter reading, tariff and load control – Direct local data exchange via hand-held unit (HHU) – ANSI Standard interface

AS 60529
Degrees of protection provided by enclosures (IP Code)

DPI Minimum AMI Functionality Specification 1.2 Victoria, September 2013

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Revision history

Version	Date	Description
V1.0	17/07/2014	Initial Version
V1.1	04/04/2016	Amended to R4.5 FW version

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